

ARAVIND KUMAR V

IoT & Machine Learning Engineer | B.Tech AI & Data Science

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PROFESSIONAL SUMMARY

Innovative AI and IoT Developer with strong expertise in Machine Learning, embedded systems, and real-time signal processing. Final year B.Tech student specializing in Artificial Intelligence and Data Science at SMVEC Puducherry with hands-on experience in developing intelligent IoT solutions for healthcare, agriculture, and assistive technologies. Proficient in Python, TensorFlow, ESP32, and Arduino.

EDUCATION

B.Tech in Artificial Intelligence & Data Science (Final Year)

Sri Manakula Vinayagar Engineering College, Puducherry

2022 – 2026

CGPA: 6.88 / 10

Higher Secondary Certificate (HSC) – Class 12 (State Board)

Amalorpavam Hr. Sec. School, Puducherry

2022

Score: 62%

TECHNICAL SKILLS

Programming:	Python, C, HTML, CSS, JavaScript (Basic)
IoT & Embedded Systems:	Arduino IDE, ESP32, Raspberry Pi, Sensor Integration, Real-Time Data Acquisition, Signal Processing
Web Development:	Flask, RESTful APIs, SQL
Machine Learning & AI:	TensorFlow, Keras, PyTorch, CNN, Image Processing
Data Science:	Pandas, NumPy, Matplotlib, Seaborn, Data Preprocessing

PROFESSIONAL EXPERIENCE

IoT + ML Developer Intern | Edusphere Solutions, Puducherry

Mar 2026 – Jun 2026

- Developed and deployed real-time IoT and Machine Learning systems using Arduino IDE, ESP32, and Python.
- Built intelligent systems integrating bio-sensors, microcontrollers, and ML models for data-driven decisions.
- Engineered Flask web applications for IoT sensor visualization, telemetry monitoring, and real-time analytics.

Python Developer Intern | Edusphere Solutions, Puducherry

Sep 2025 – Dec 2025

- Developed Python applications and automated scripts for structured data processing and analytics workflows.
- Implemented data preprocessing pipelines and feature engineering techniques for machine learning models.

KEY PROJECTS

- Real-Time ECG, EMG, EOG Signal Acquisition & Anomaly Detection:** Developed end-to-end bio-signal capture system using BioAmp EXG + ESP32. Built TensorFlow deep learning models classifying arrhythmias and muscular anomalies with live alerts.
- EOG Wheelchair with Fall Detection:** Hands-free assistive wheelchair guided by eye movement signals (EOG) with MPU6050 IMU fall detection and automated GSM caregiver SMS alerts.
- EMG-Based Wheelchair Control System:** Machine learning classification of muscle contraction signals to drive directional wheelchair motion for motor-impaired individuals.
- Smart Agriculture AI & Threat Detection:** CNN-powered foliar leaf disease detection and wildlife intrusion monitoring with automated sensor-based irrigation control.
- Airbag Collision Detection & Safety:** MPU6050 G-force shock detection triggering servo airbag simulation & emergency GSM distress SMS.

CERTIFICATIONS & ACHIEVEMENTS

- Introduction to Machine Learning & Python for Data Science
- AWS Academy Graduate – AWS Academy Cloud Foundations
- Datathon Participant (SRM University, Feb 2025) – National-level Data Science Hackathon